Chapter 5 exercises

DS05-01 • Traversals and height

The root is 8. Its left child 3 has children 1 and 6. Its right child 12 has left child 10. Find preorder, inorder, postorder, level order, tree height, every node's depth, and maximum level width. State DFS and BFS auxiliary-space bounds.

DS05-02 • BST insertion and two-child deletion

Insert 11 into the BST in DS05-01, then delete 8 using its successor. Draw the result and identify the physically removed node position. Explain the semantic issue if external handles identify nodes and deletion moves values.

DS05-03 • Heap updates

Start with min-heap `[2,5,4,9,7,8]`. Insert 1, then delete-min, listing every swap. Do not replace heap operations with sorting the whole array.

DS05-04 • Prove linear heap construction

Explain why "n sift-downs, each O(log n), hence Θ(n log n)" is not a tight-bound proof. Prove Θ(n) by summing costs over heights. State the worst-case cost of construction by individual insertions.

DS05-05 • From ordered sources to a balanced tree

Repeatable ordered sources are [2,7,17],[5,7,13],[11,19], with value 1 per record. Aggregate equal keys into occurrence counts. Design construction without repeated ordinary BST insertion; show every key/value, parent, and height. Separate merging and construction costs when source count k grows. Trace successor(7) and the maximum key’ s successor.

References and attribution

Ellis Horowitz, Sartaj Sahni, Dinesh Mehta. Fundamentals of Data Structures in C++, 2nd Edition.

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https://github.com/xinhi6716/computer-science-self-study